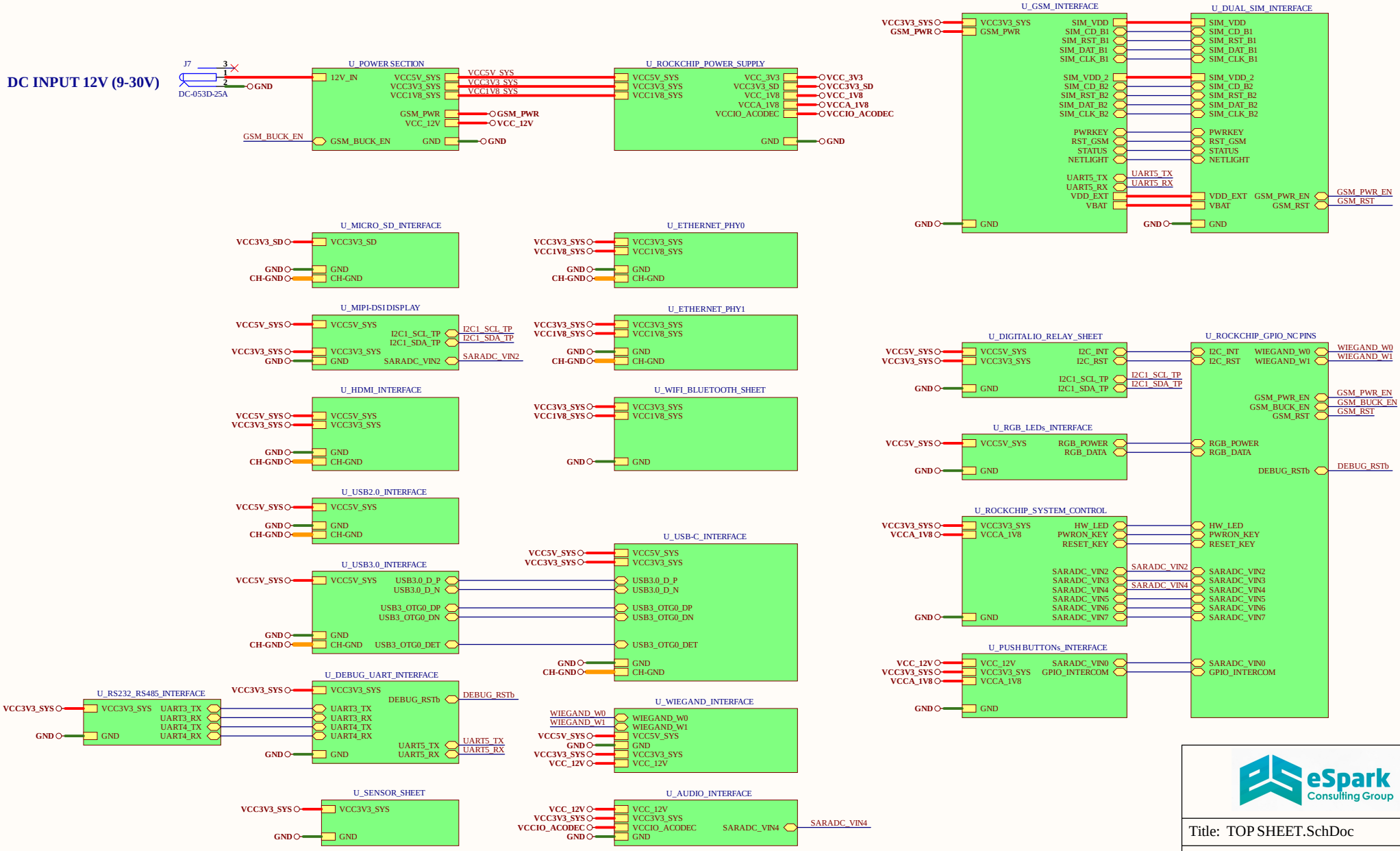
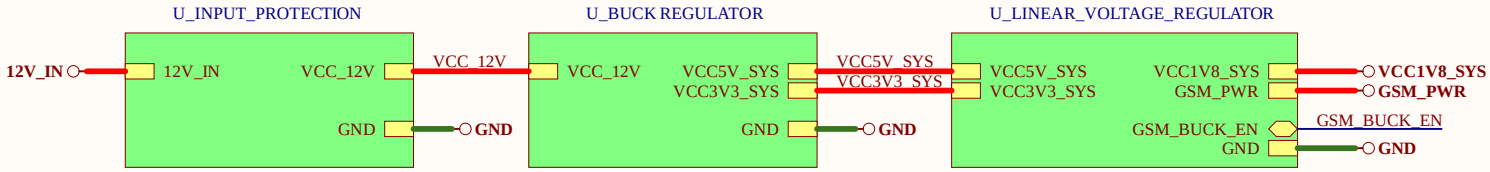
TOP SHEET

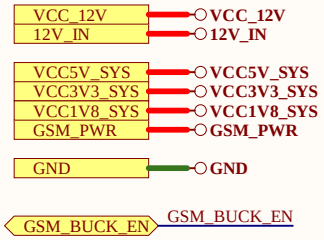


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Title: TOP SHEET.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 27

POWER SECTION

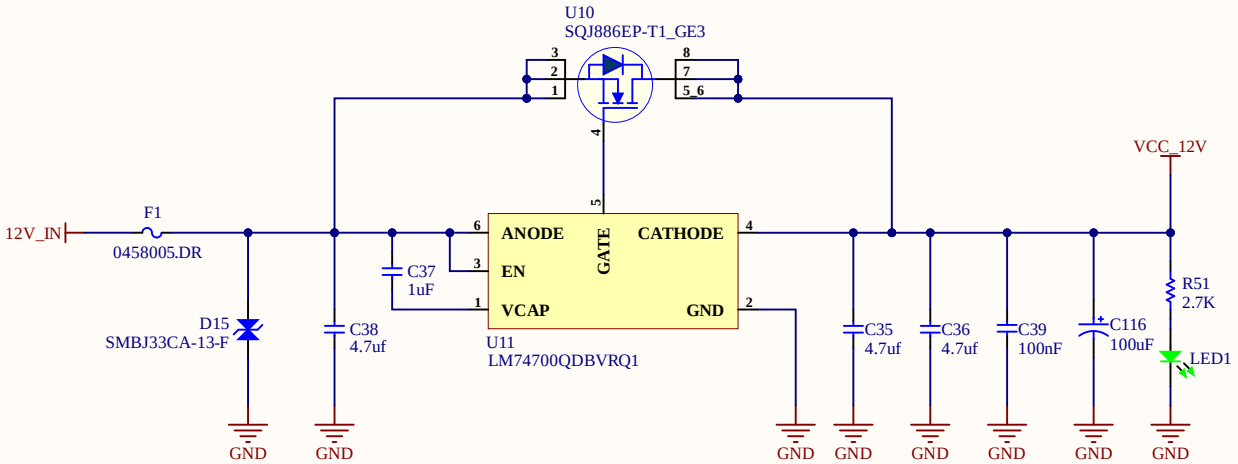




eSpark Consulting Group

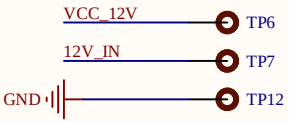
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File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 27

INPUT PROTECTION CIRCUIT



REVERSE POLARITY PROTECTION

TEST POINTs

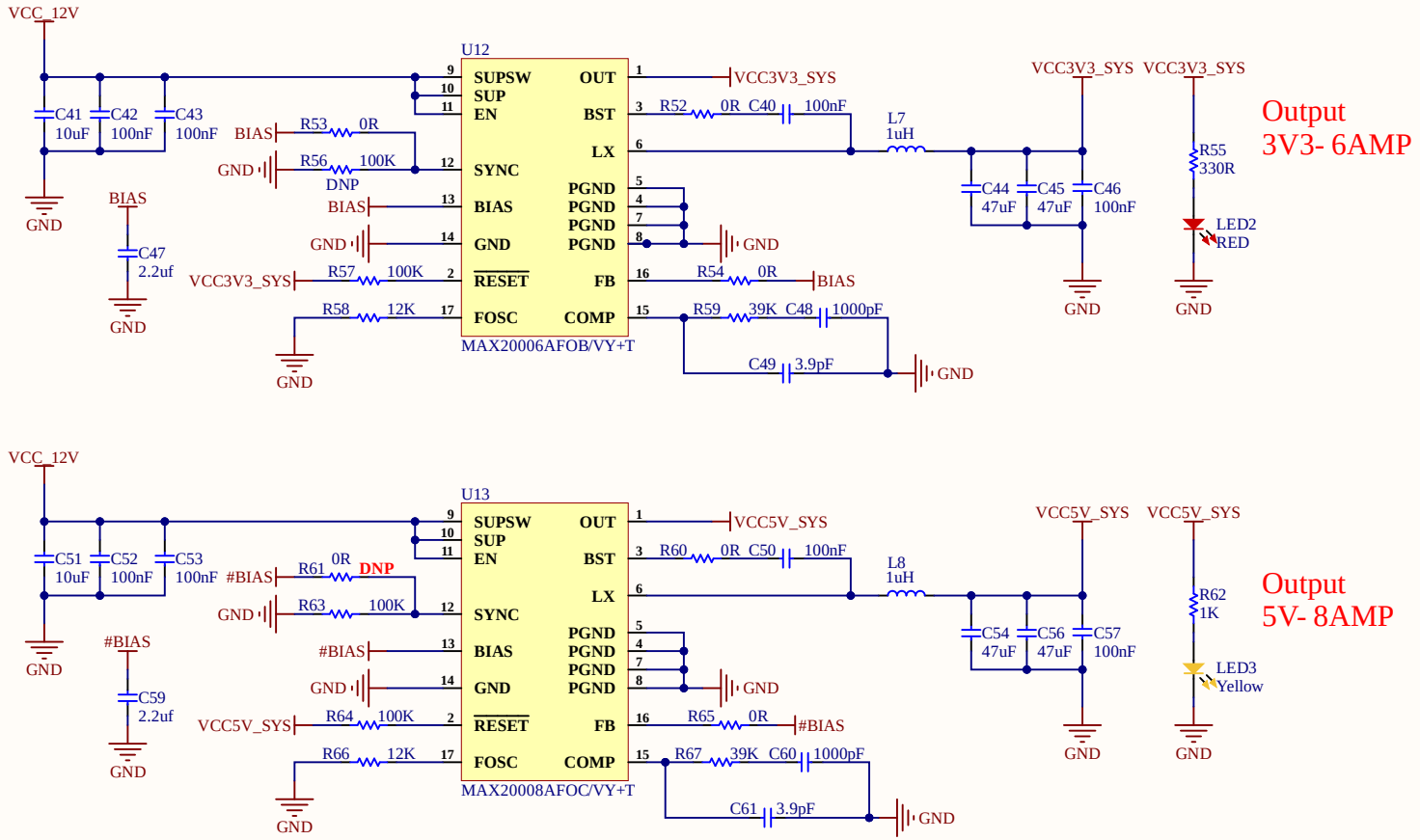




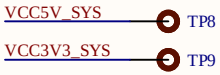
eSpark Consulting Group

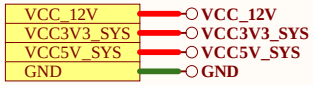
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File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 27

BUCK REGULATOR



TEST POINTs



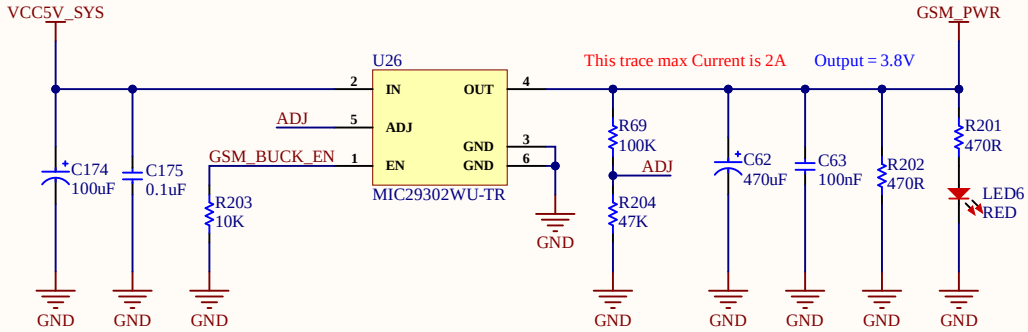


eSpark Consulting Group

Title: BUCK REGULATOR.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 27

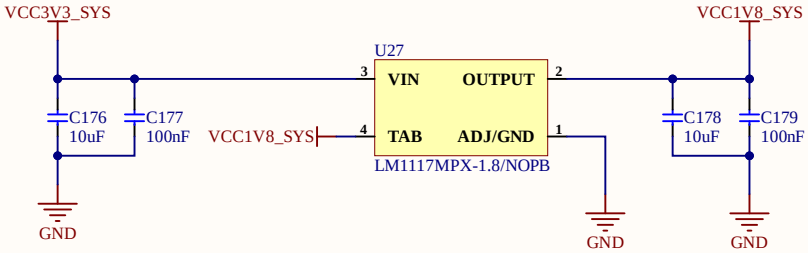
LINEAR_VOLTAGE_REGULATOR

LDO FOR GSM POWER



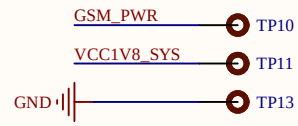
Output
3V8- 3AMP

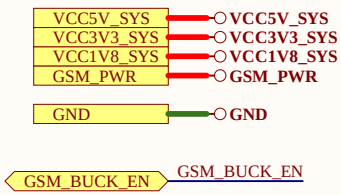
1V8 LDO



Output
1V8- 0.8AMP

TEST POINTs



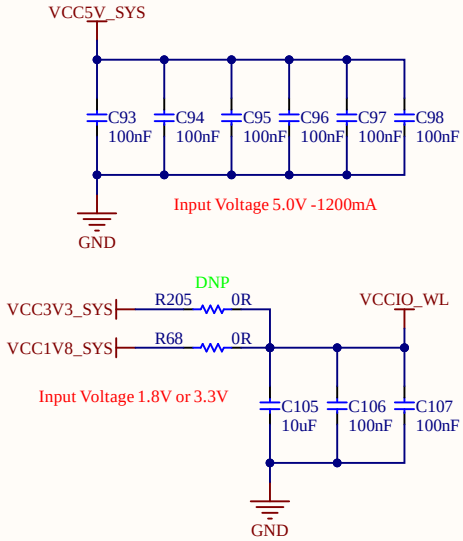


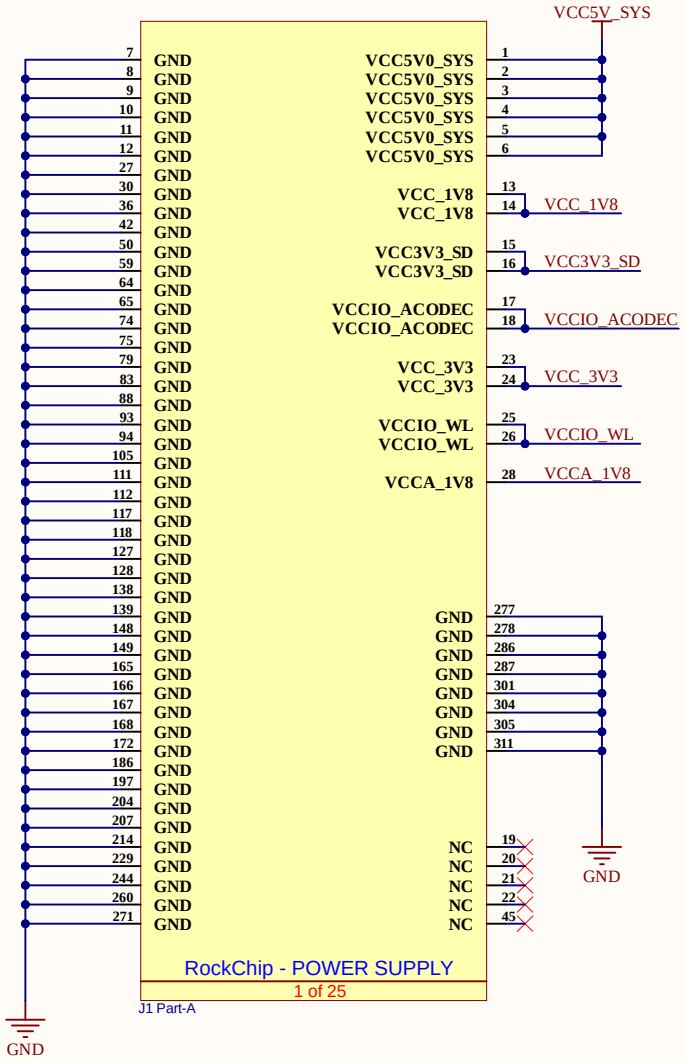
eSpark Consulting Group

Title: LINEAR_VOLTAGE_REG.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 24/05/2024	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 27

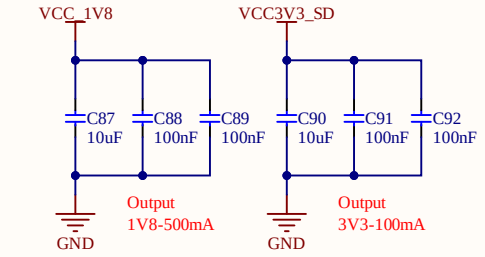
ROCKCHIP POWER SUPPLY

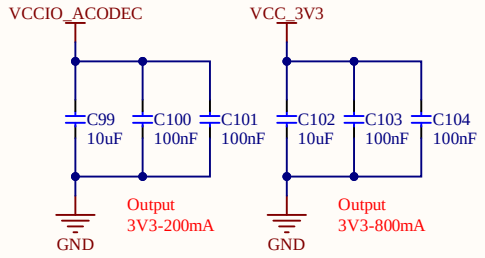
DECOUPLING CAPACITORs
PLACE CLOSE TO EACH PIN

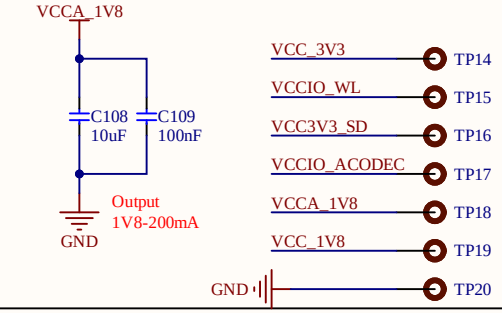




DECOUPLING CAPACITORs
PLACE CLOSE TO EACH PIN







VCC3V3_SYS — VCC3V3_SYS
VCC5V_SYS — VCC5V_SYS
GND — GND

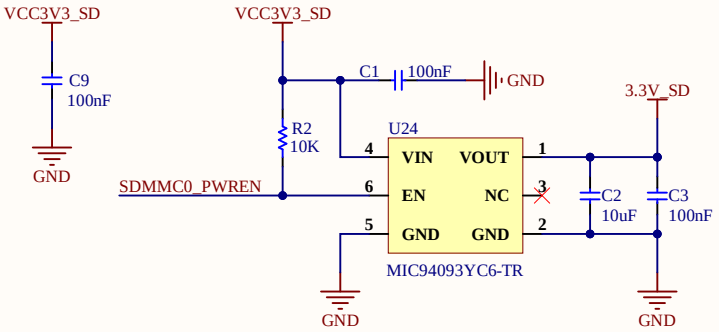
VCC_3V3 — VCC_3V3
VCC_1V8 — VCC_1V8
VCCA_1V8 — VCCA_1V8
VCC3V3_SD — VCC3V3_SD
VCC1V8_SYS — VCC1V8_SYS

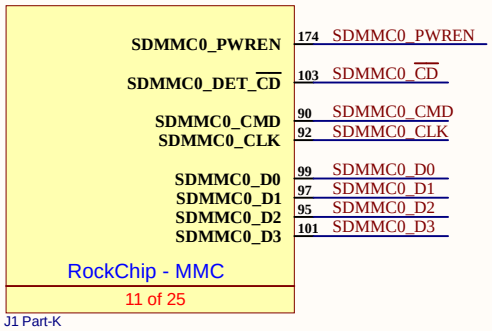
VCCIO_ACODEC — VCCIO_ACODEC

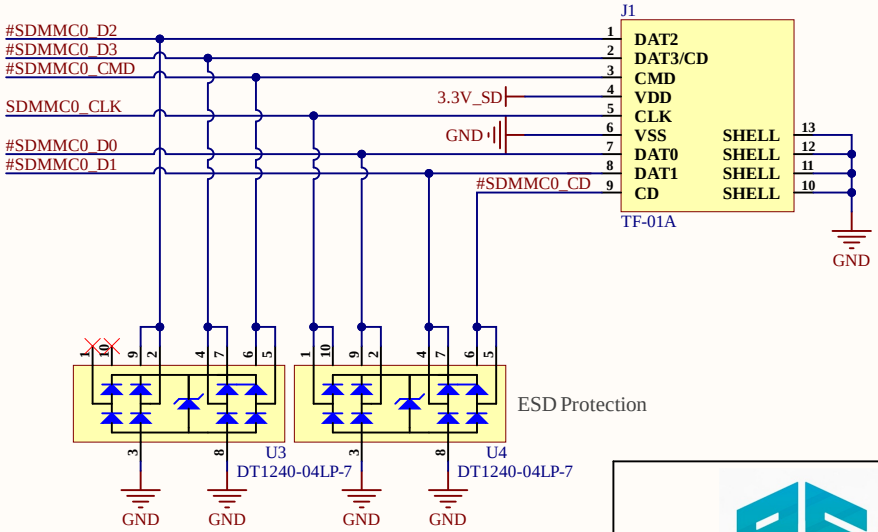
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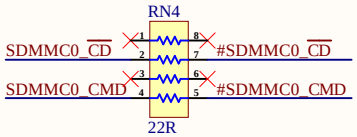
Title: ROCKCHIP_PWR_SUPPLY.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 27

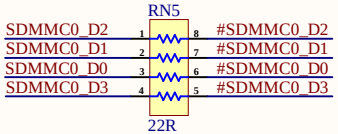
MICRO SD CARD INTERFACE

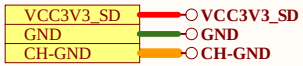








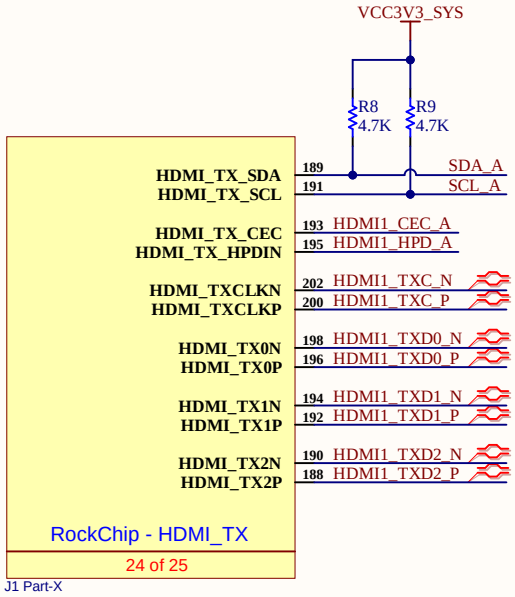


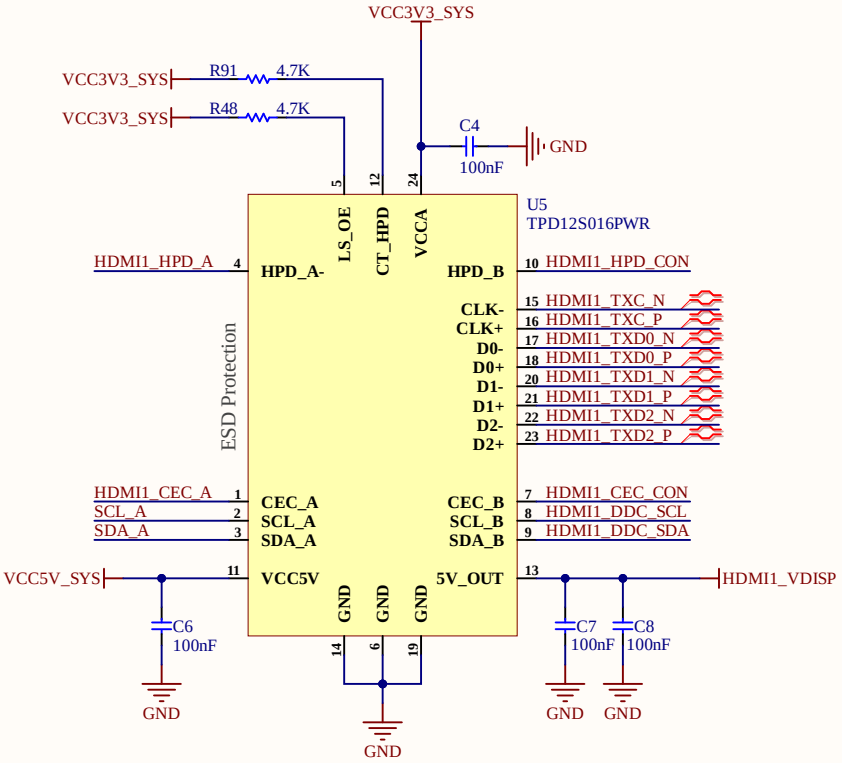


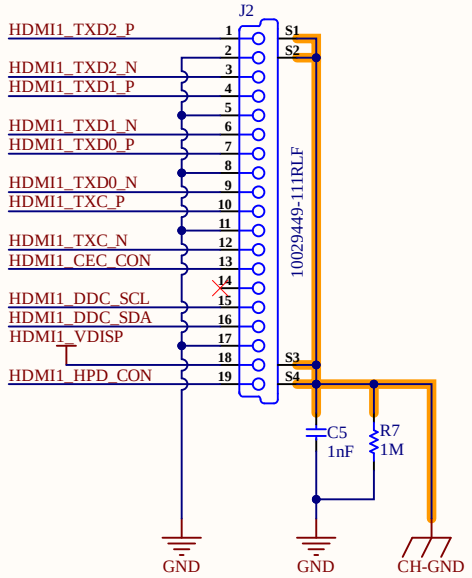
eSpark Consulting Group

Title: MICRO_SD_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page	Num: 7
Drawn By: Huzaifa Ali	Page	Total: 27

HDMI INTERFACE

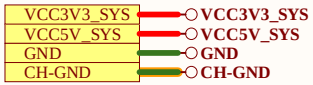






TEST POINTs

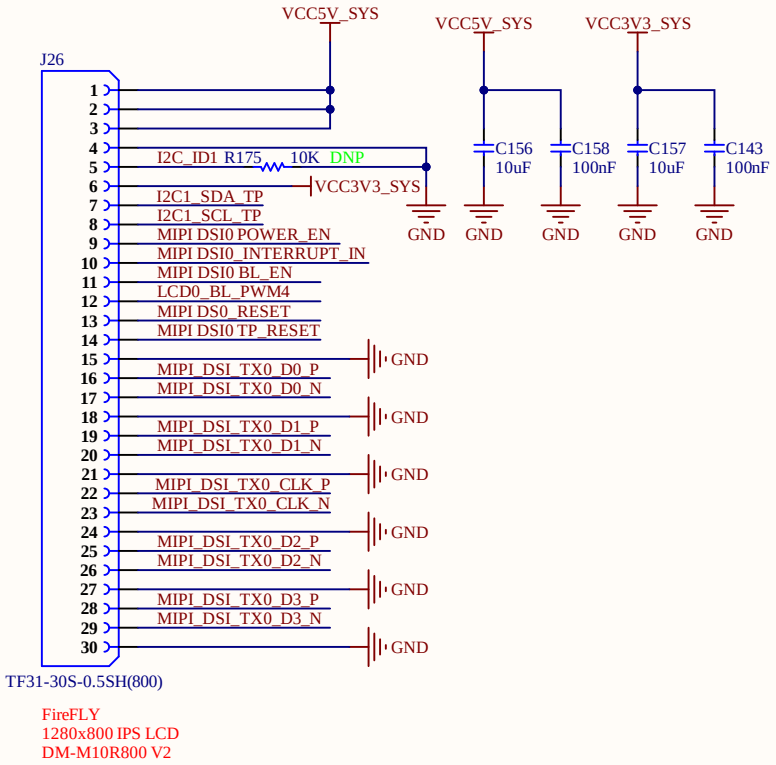


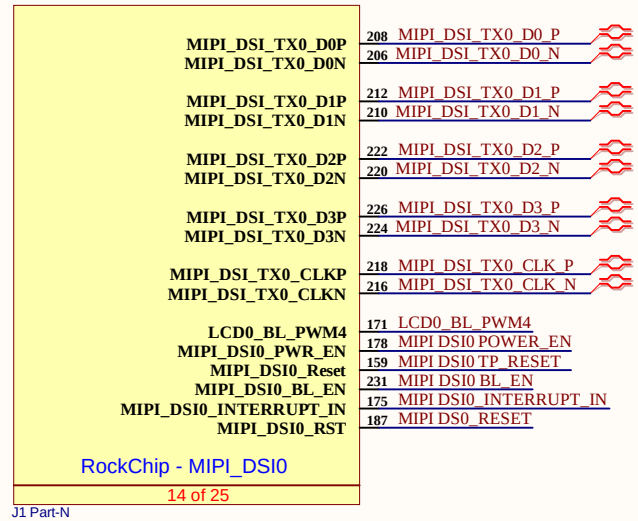


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Title: HDMI_INTERFACE.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 8
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MIPI-DSI DISPLAY





SARADC_VIN2_LCD_ID R116 0R I2C_ID1

I2C1_SDA_TP — I2C1_SDA_TP
I2C1_SCL_TP — I2C1_SCL_TP

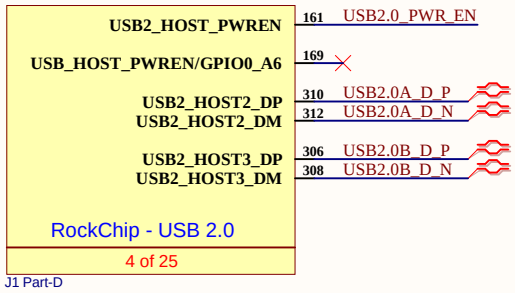
SARADC_VIN — SARADC_VIN2_LCD_ID

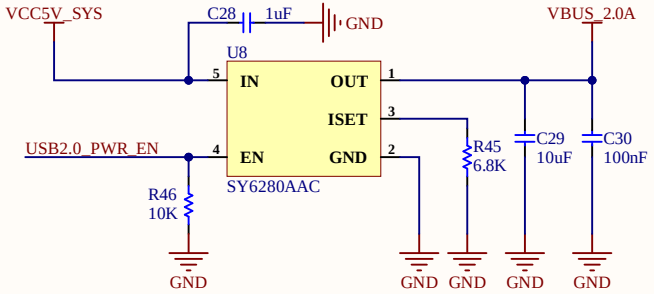
VCC3V3_SYS — VCC3V3_SYS
VCC5V_SYS — VCC5V_SYS
GND — GND

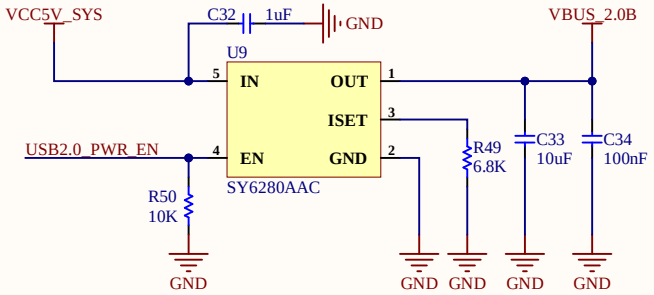
eSpark Consulting Group

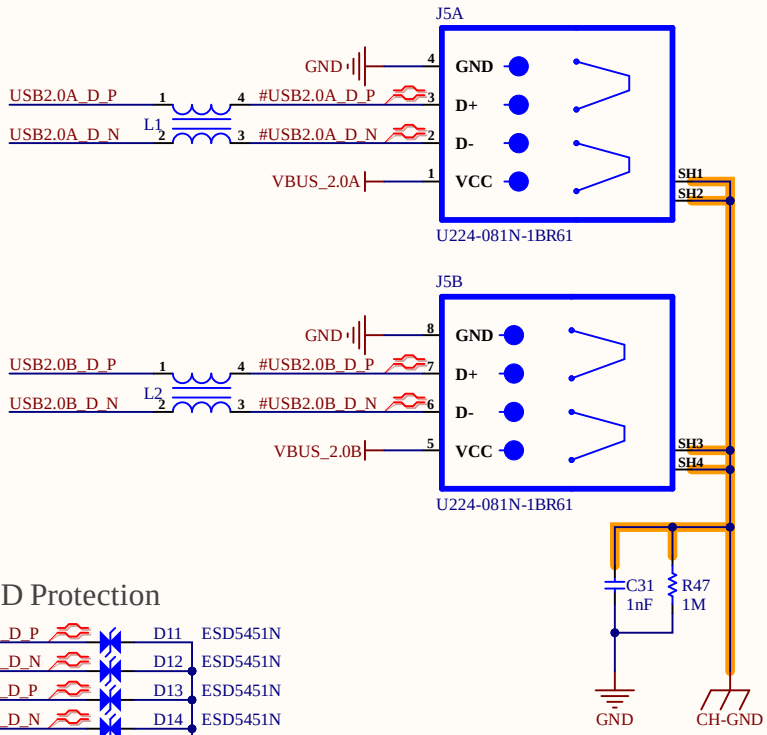
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File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 27

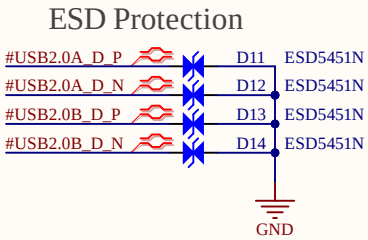
USB2.0 INTERFACE

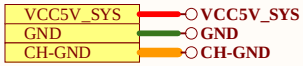








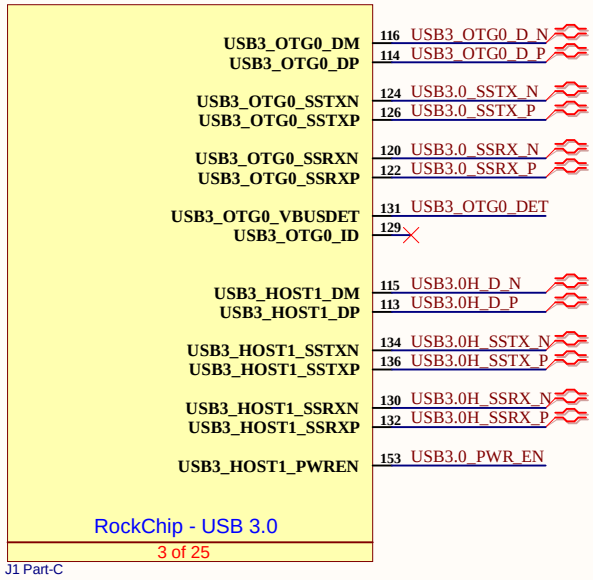


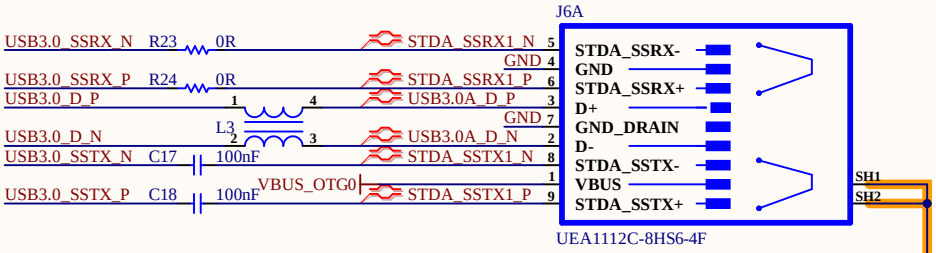


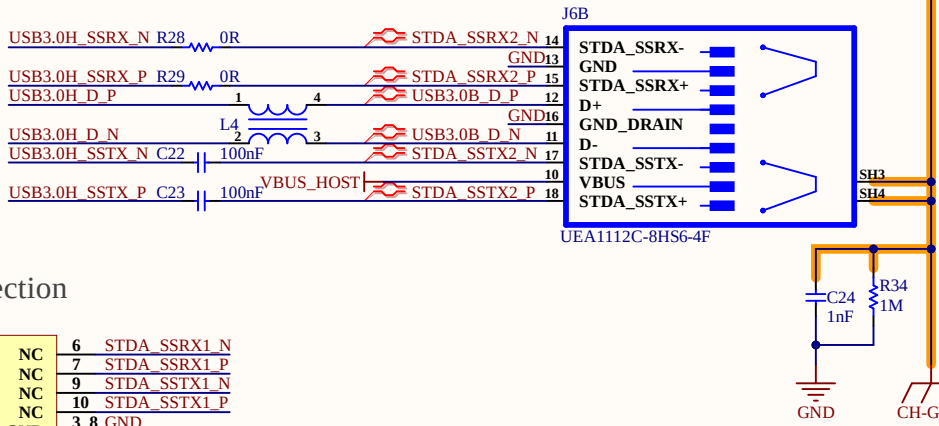
eSpark Consulting Group

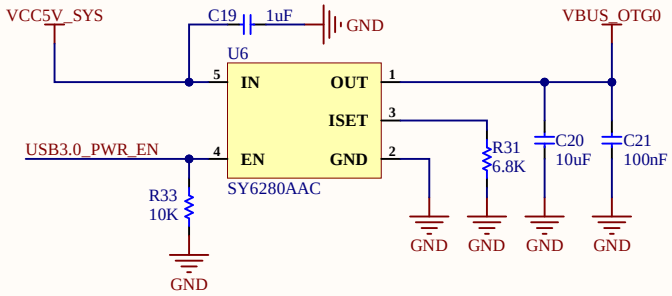
Title: USB2.0_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 20/05/2024	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 27

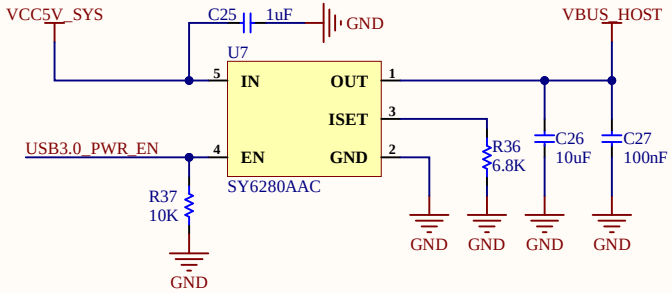
USB3.0 INTERFACE



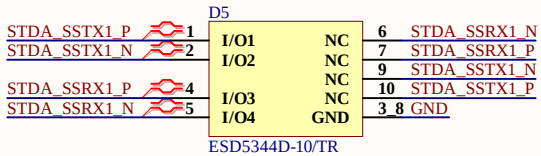


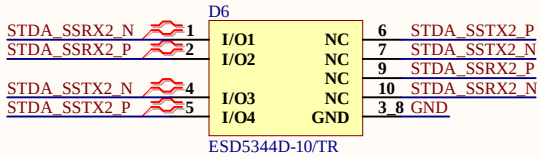


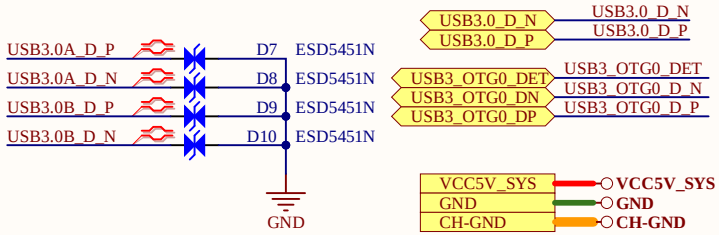




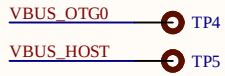
ESD Protection







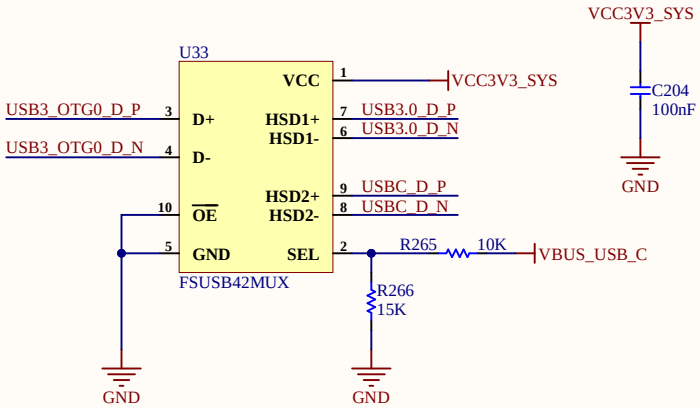
TEST POINTs

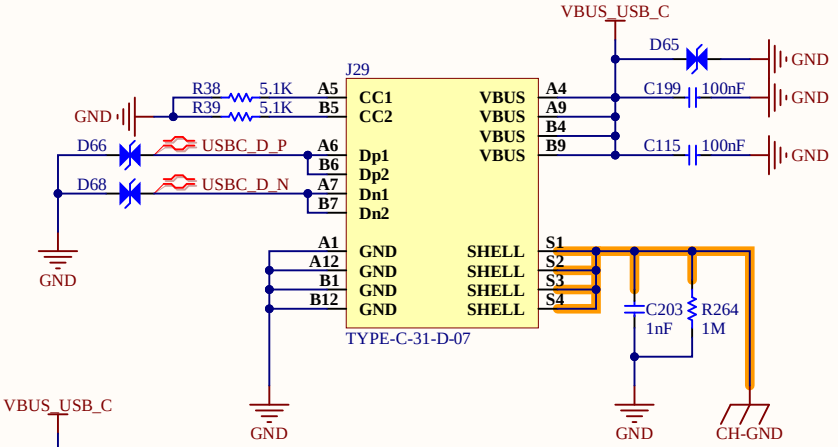


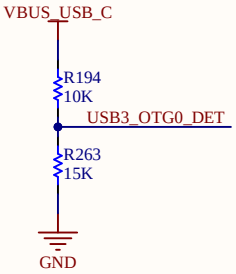
eSpark Consulting Group

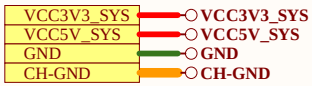
Title: USB3.0_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page Num:	11
Drawn By: Huzaifa Ali	Page Total:	27

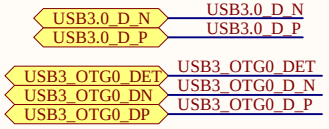
USB-C_INTERFACE







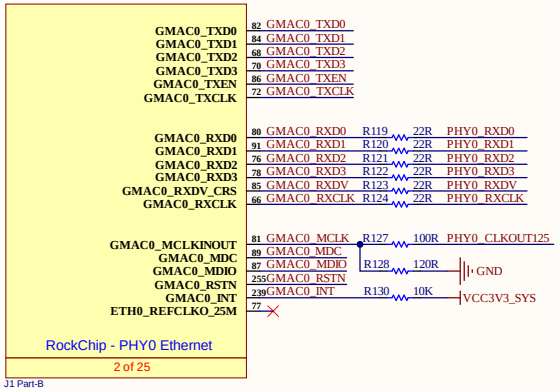


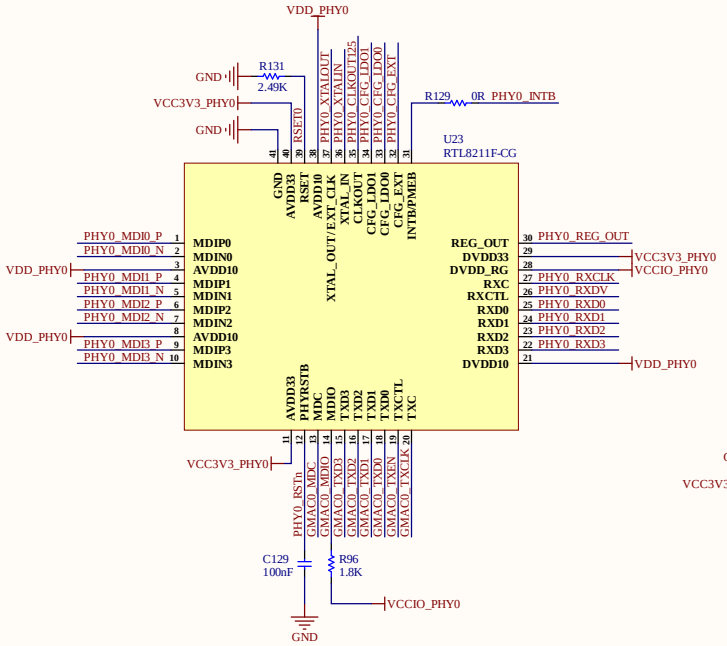


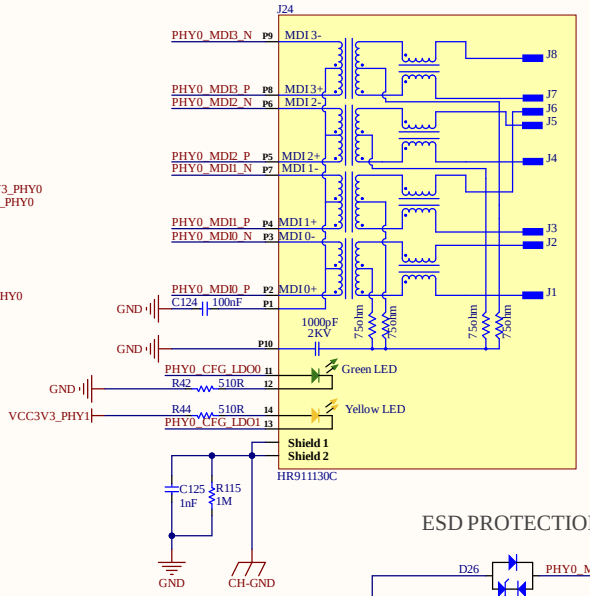
eSpark Consulting Group

Title: USB-C_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page Num:	12
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ETHERNET_PHY0_INTERFACE







PLL MODE

PHY0_RXD2 R97 4.7K VCCIO_PHY0

Pull-up to disable PLL

TXC/RXC DELAY CONFIG

PHY0_RXD0 R99 4.7K GND

PHY0_RXD1 R132 4.7K GND

Pull-up for additional 2ns Delay

PHY0 ADDRESS CONFIG

PHY0_RXD3 R98 4.7K VCCIO_PHY0

PHY0_RXCLK R100 4.7K GND

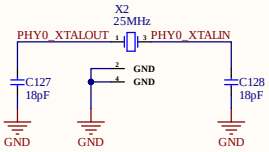
PHY0_RXDV R101 4.7K GND

PHY0_IO Voltage Config

PHY0_CFG_EXT R133 4.7K VCC3V3_PHY0

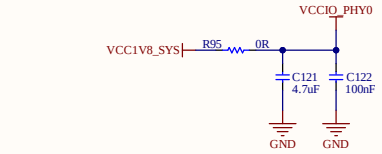
PHY0_CFG_LDO1 R134 4.7K VCC3V3_PHY0

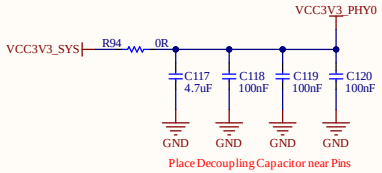
PHY0_CFG_LDO0 R135 4.7K GND

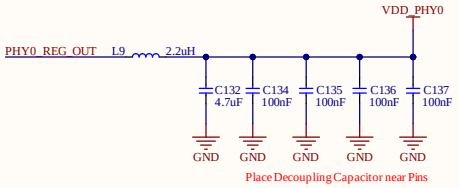


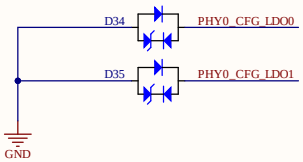
GMAC0_INT R93 0R PHY0_INTB

GMAC0_RSTN R125 0R PHY0_RSTn

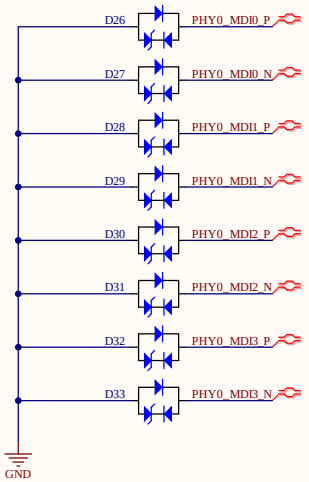








ESD PROTECTION

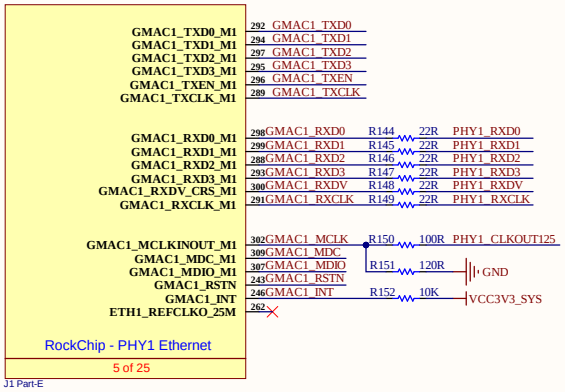


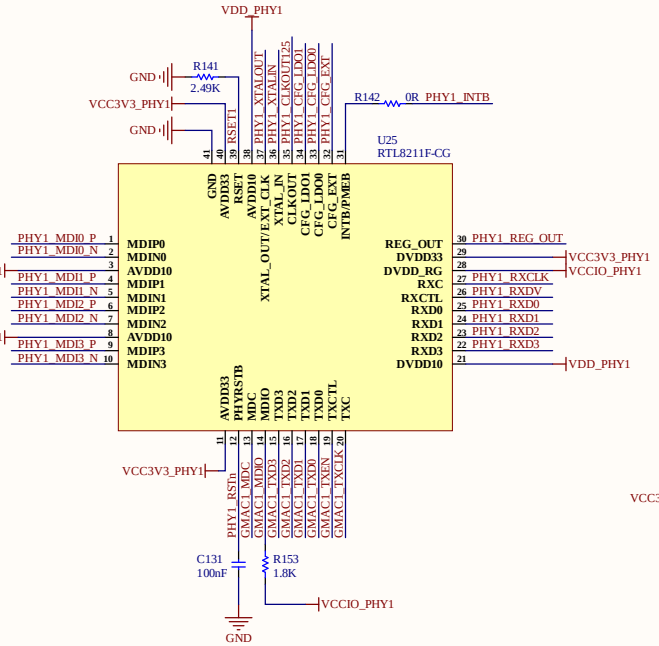
VCC3V3_SYS	VCC3V3_SYS
VCC1V8_SYS	VCC1V8_SYS
GND	GND
CH-GND	CH-GND

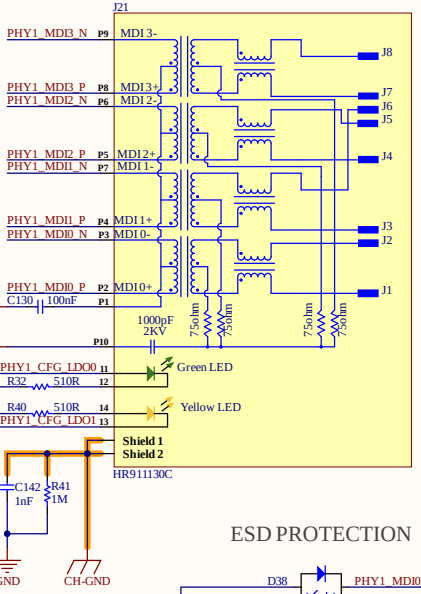
eSpark Consulting Group

Title: ETHERNET_PHY0.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
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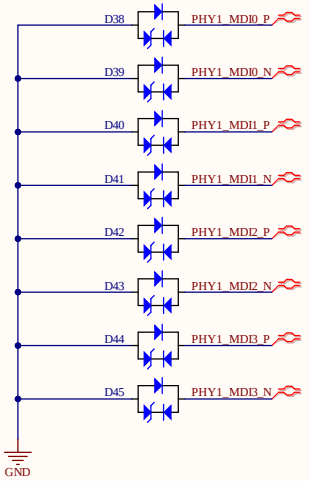
ETHERNET_PHY1_INTERFACE







ESD PROTECTION



PLL MODE

PHY1_RXD2 R156 4.7K VCCIO_PHY1

Pull-up to disable PLL

TXC/RXC DELAY CONFIG

PHY1_RXD0 R160 4.7K GND

PHY1_RXD1 R161 4.7K GND

Pull-up for additional 2ns Delay

PHY1 ADDRESS CONFIG

PHY1_RXD3 R165 4.7K VCCIO_PHY1

PHY1_RXCLK R166 4.7K GND

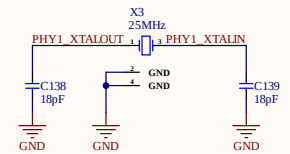
PHY1_RXDV R167 4.7K GND

PHY1_IO Voltage Config

PHY1_CFG_EXT R170 4.7K VCC3V3_PHY1

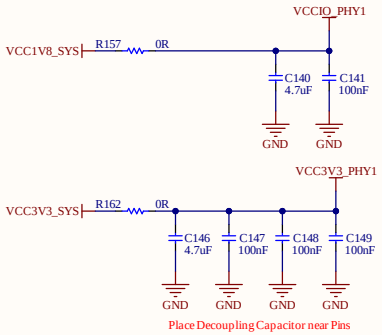
PHY1_CFG_LDO1 R171 4.7K VCC3V3_PHY1

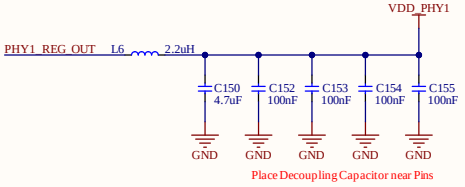
PHY1_CFG_LDO0 R172 4.7K GND

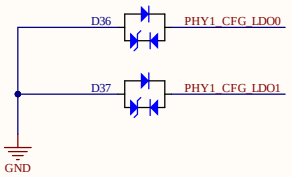


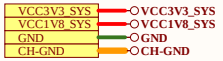
GMAC1_INT R126 0R PHY1_INTB

GMAC1_RSTN R136 0R PHY1_RSTn



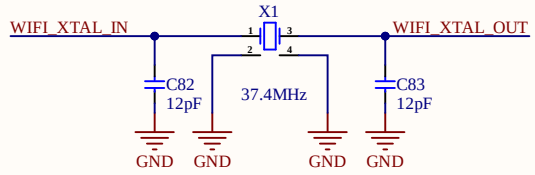


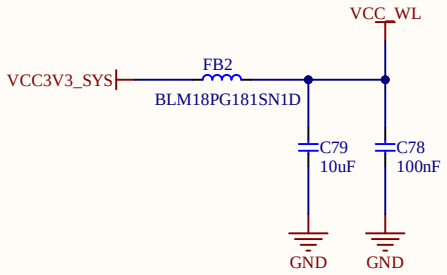




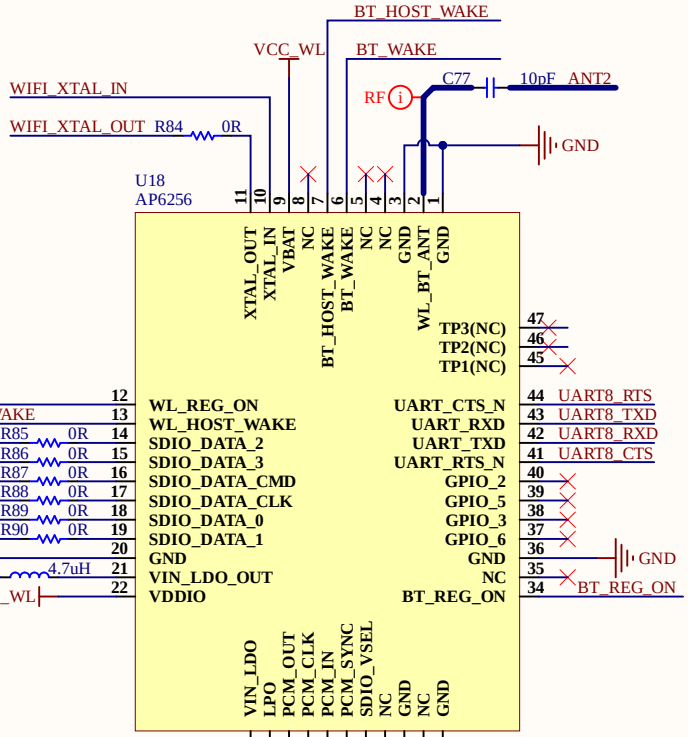
Title: ETHERNET_PHY1.SchDoc	
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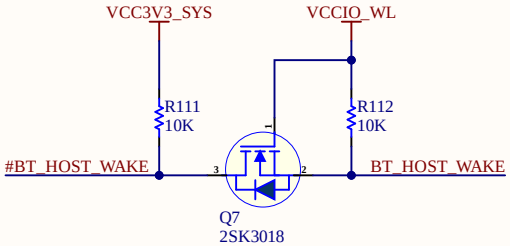
WIFI_BLUETOOTH_INTERFACE

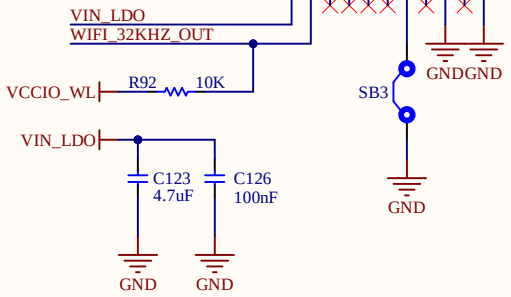


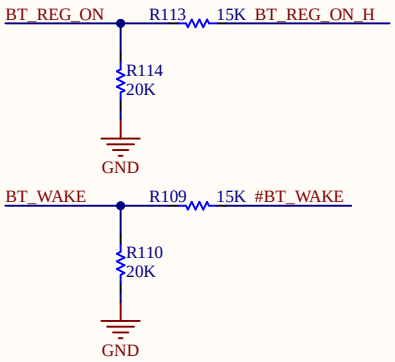


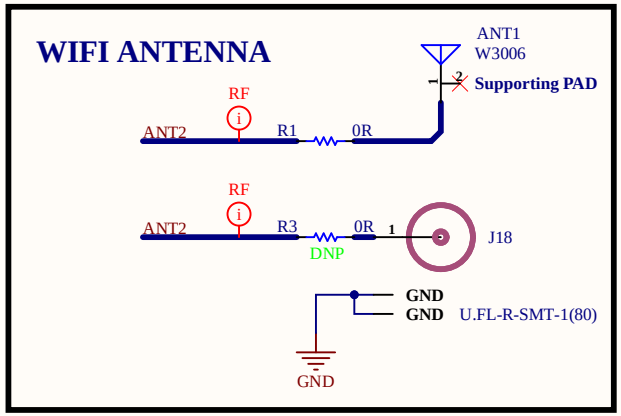
Dual Band WiFi

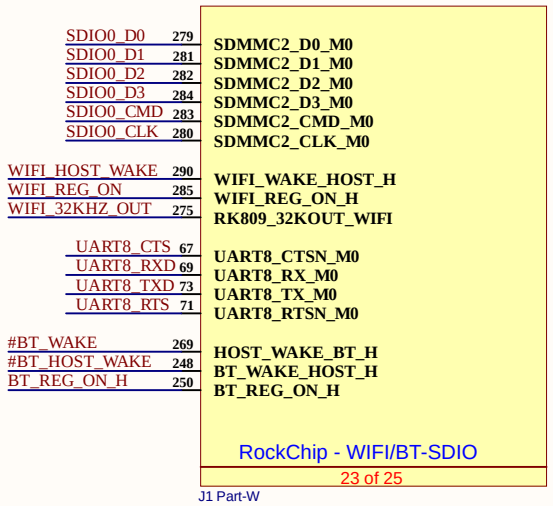


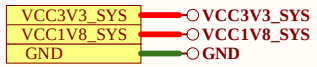






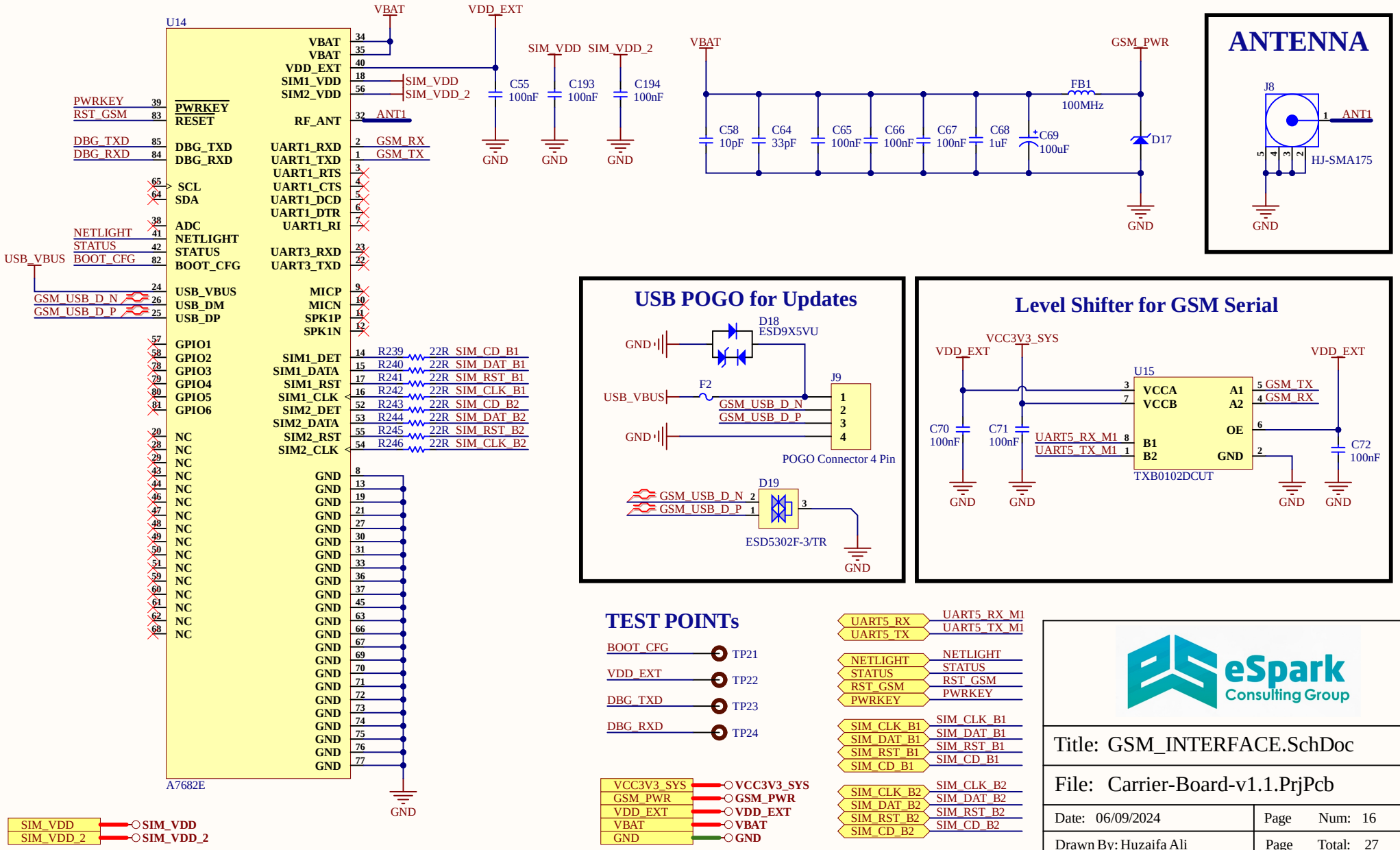






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Date: 09/09/2024	Page Num:	15
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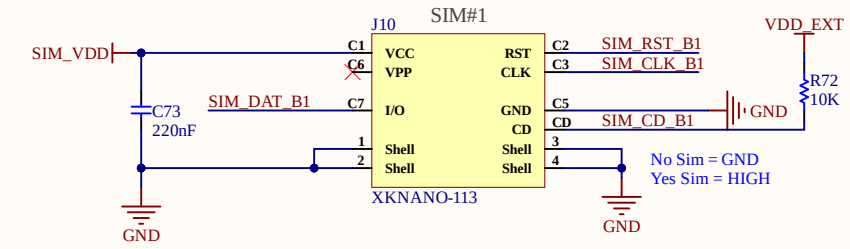
GSM_INTERFACE

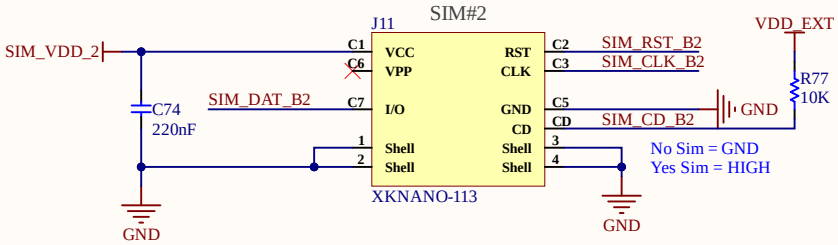


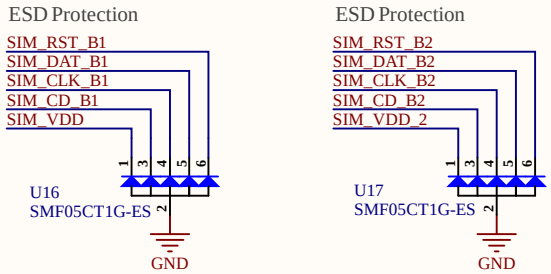
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File:	Carrier-Board-v1.1.PrjPcb
Date: 06/09/2024	Page Num: 16
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DUAL_SIM_INTERFACE

DUALSIM WITH MULTIPLEXER

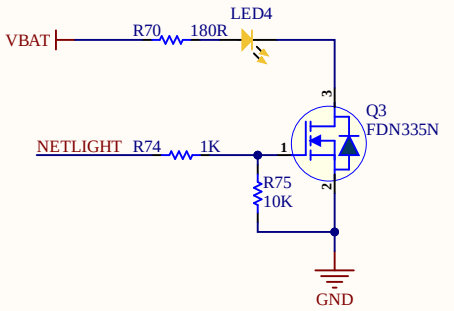


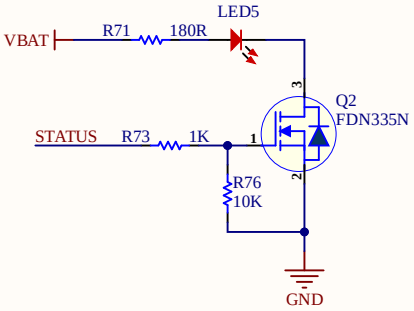


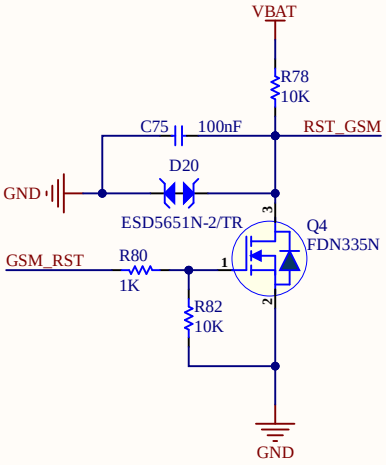


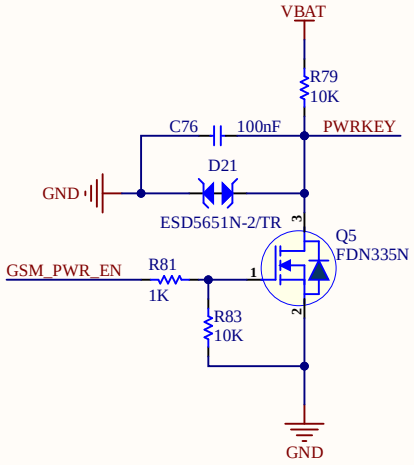
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

STATUS LEDs



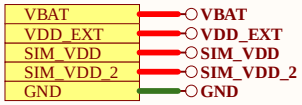


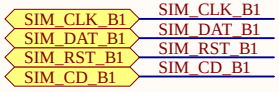


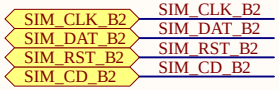










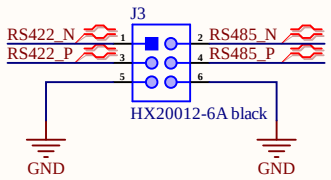


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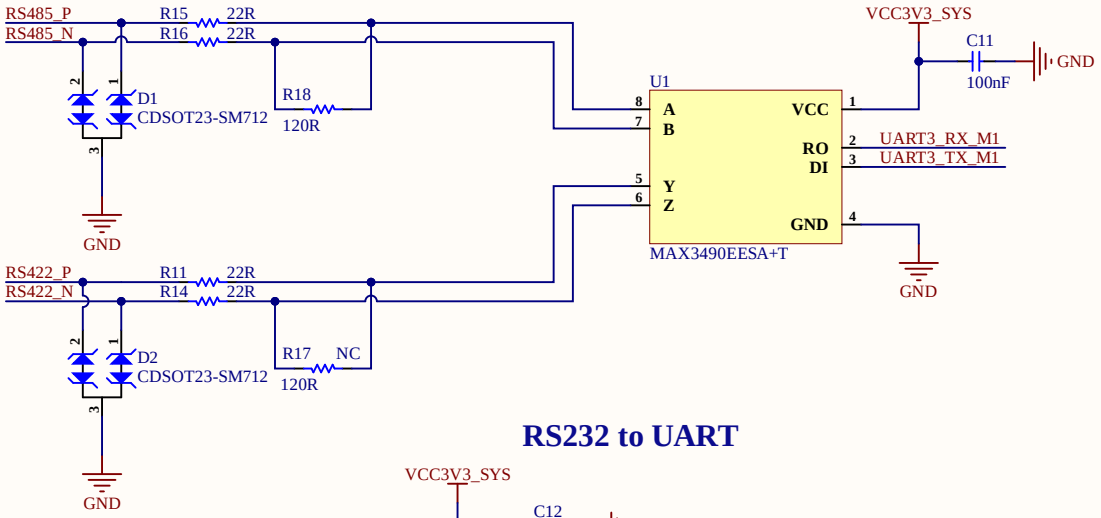
Title: DUAL_SIM_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page	Num: 17
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RS232_RS485_INTERFACE

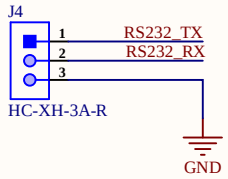
PORT for RS485



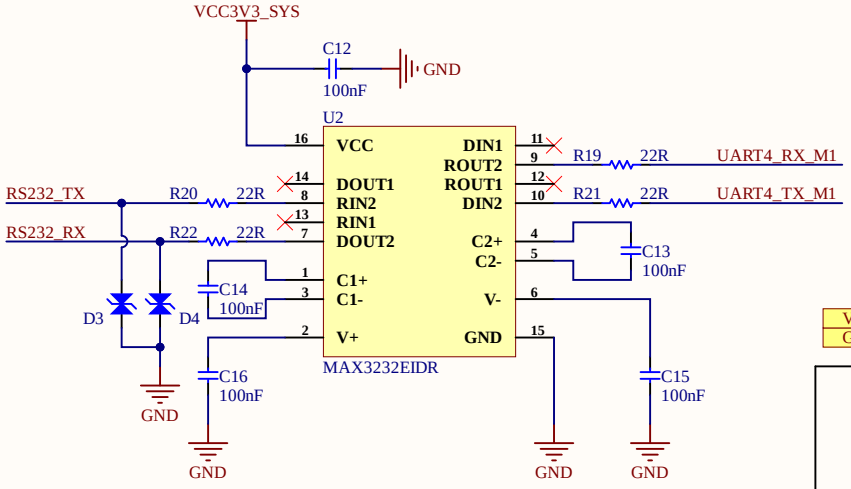
RS485 to UART



PORT for RS232



RS232 to UART



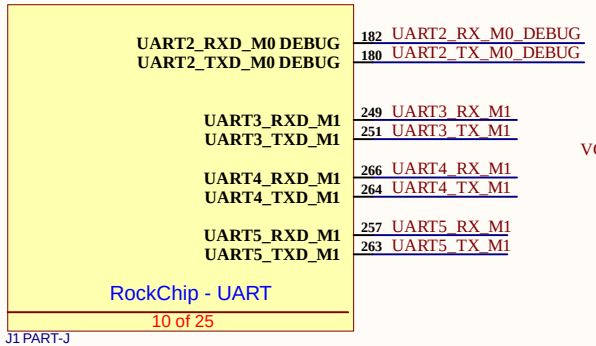


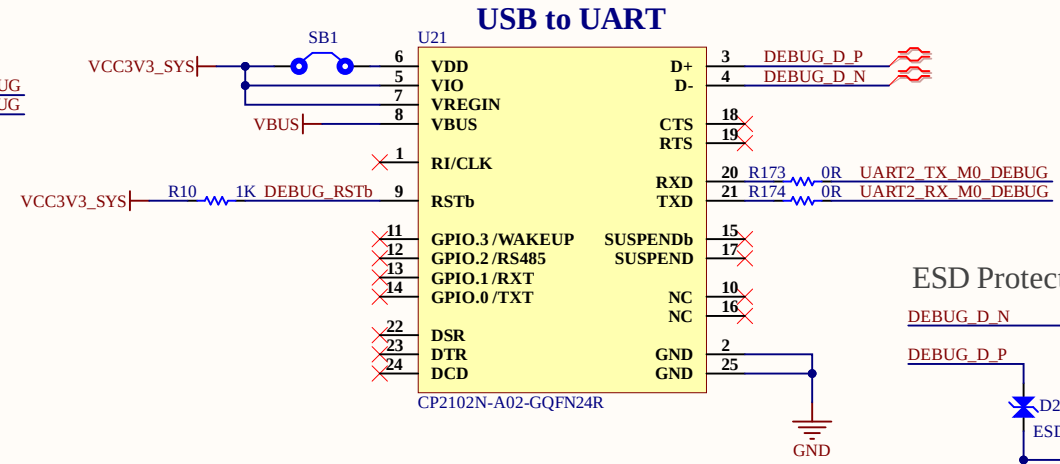


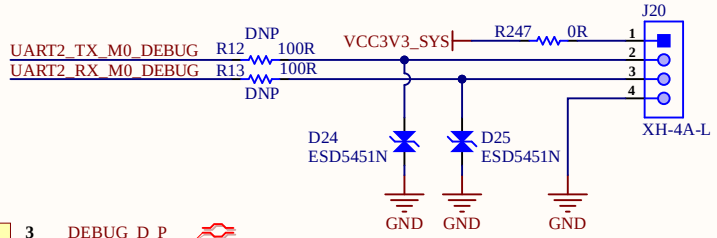
eSpark Consulting Group

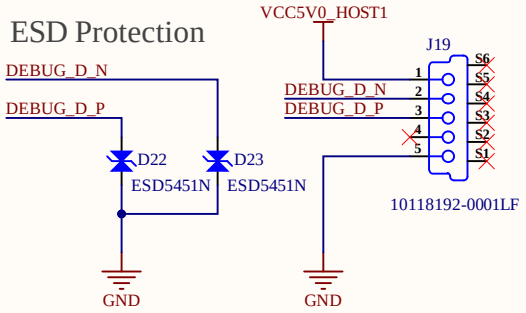
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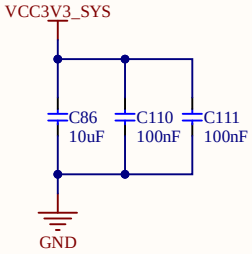
DEBUG_UART_INTERFACE

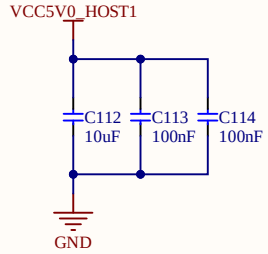


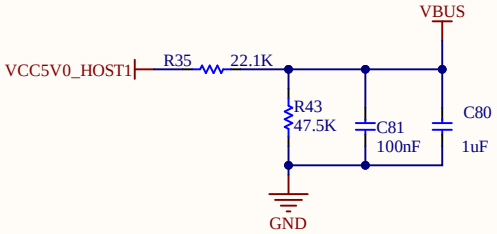


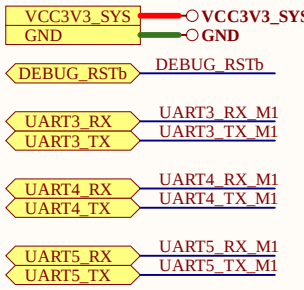










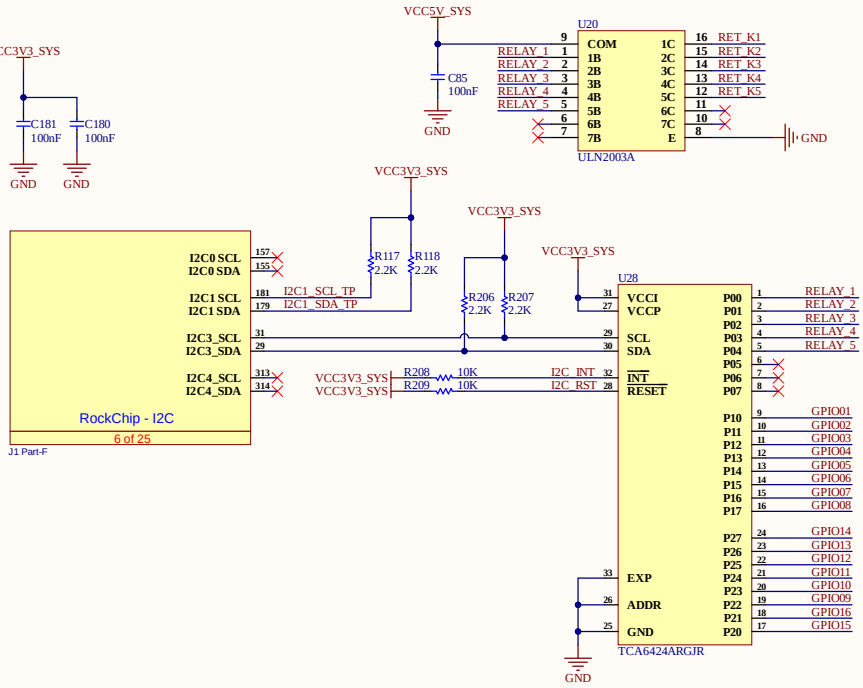


eSpark Consulting Group

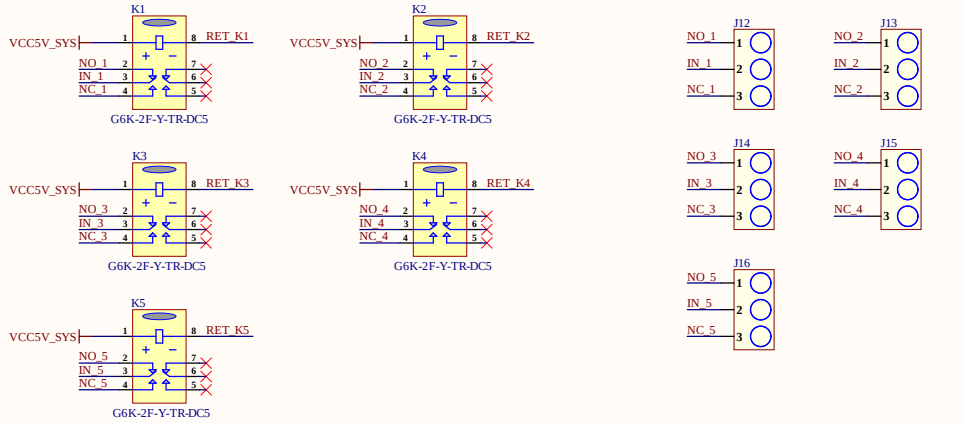
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File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 19
Drawn By: Huzaifa Ali	Page	Total: 27

DIGITAL IOs_RELAY_INTERFACE

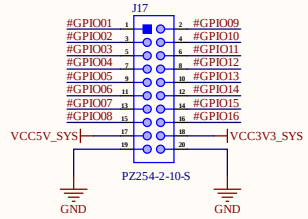
RELAY WITH DRIVER



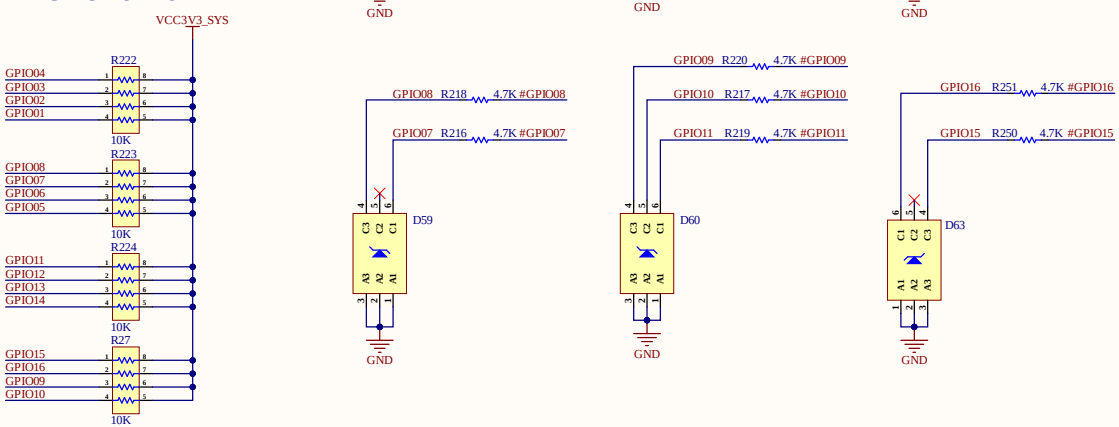
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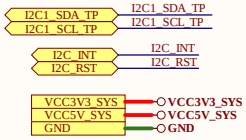


DIGITAL I/O



GPIO PULLUP

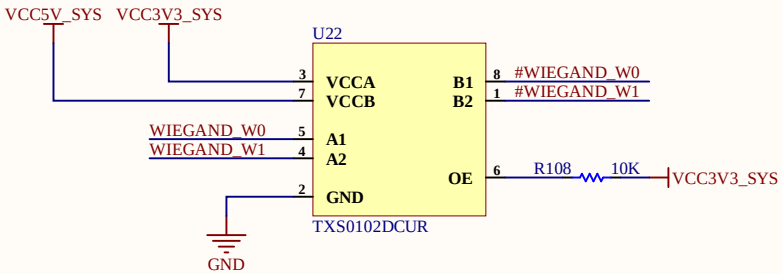


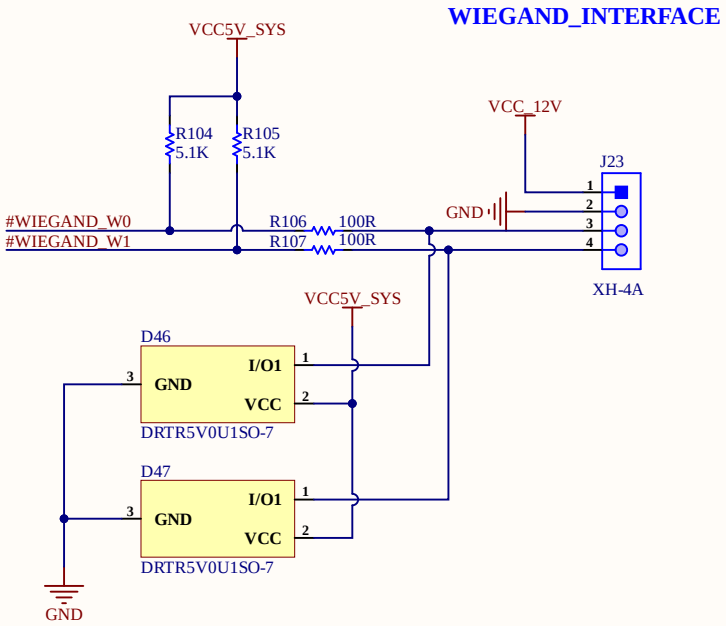


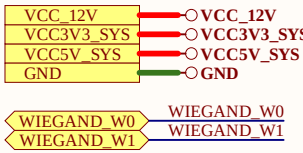
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Title: DIGITAL IO_RELAY_SHEET.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 06/09/2024	Page	Num: 20
Drawn By: Huzaifa Ali	Page	Total: 27

WIEGAND_INTERFACE



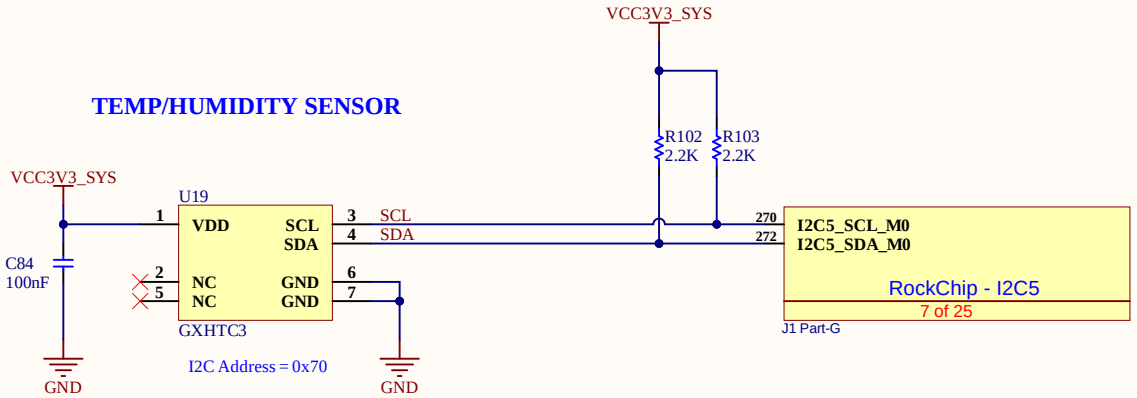




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Title: WIEGAND_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 21
Drawn By: Huzaifa Ali	Page	Total: 27

TEMP/HUMIDITY SENSOR_INTERFACE

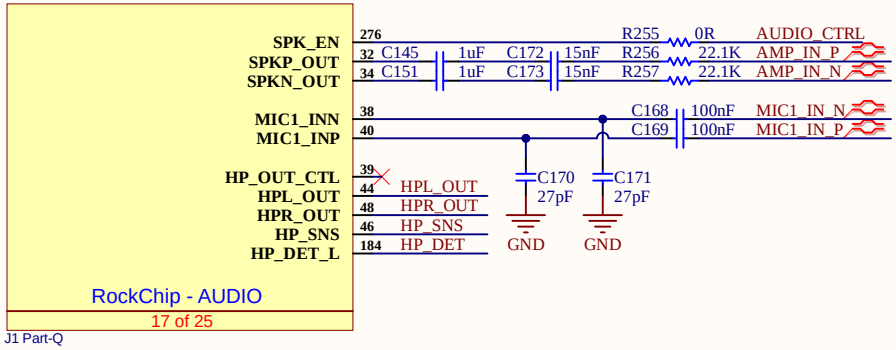


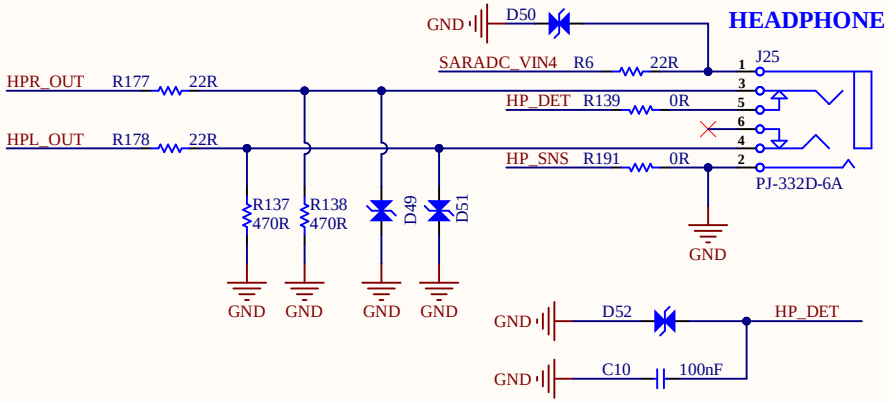


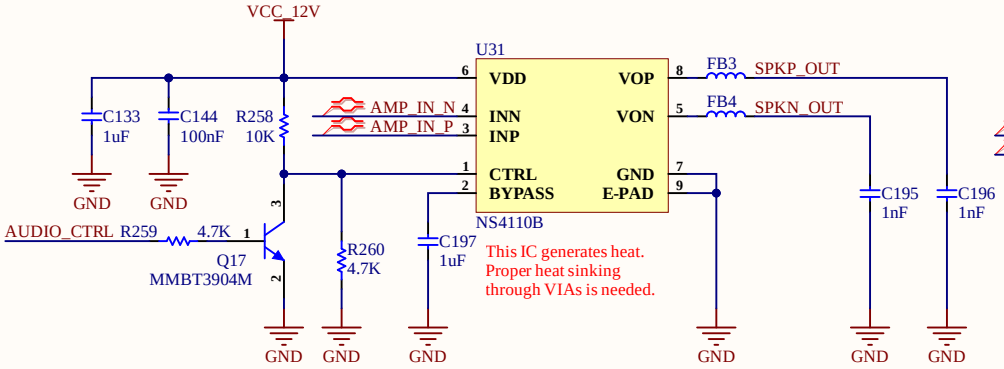
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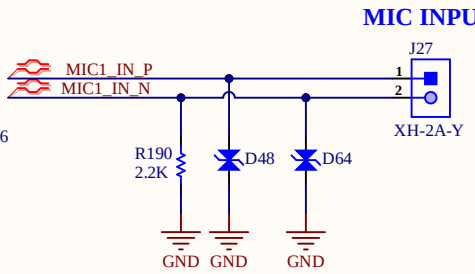
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File: Carrier-Board-v1.1.PrjPcb	
Date: 10/05/2024	Page Num: 22
Drawn By: Huzaifa Ali	Page Total: 27

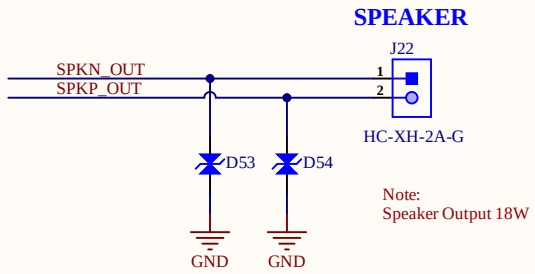
AUDIO_INTERFACE

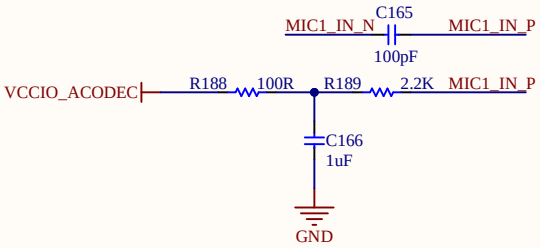


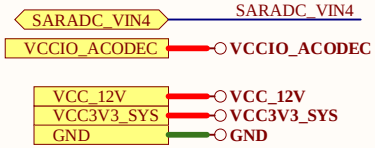










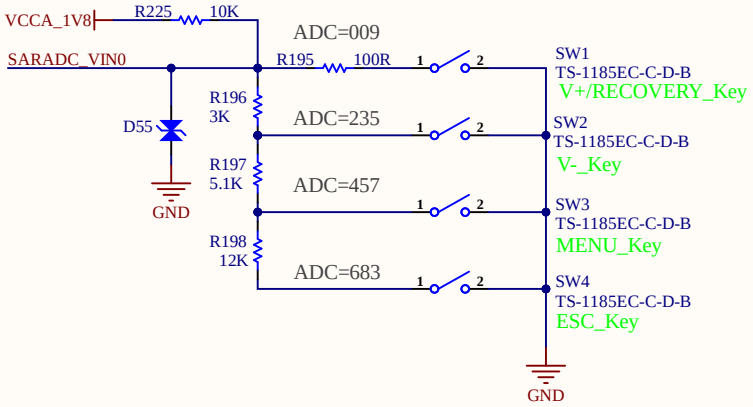


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File: Carrier-Board-v1.1.PrjPcb		
Date: 09/09/2024	Page Num:	23
Drawn By: Huzaifa Ali	Page Total:	27

PUSH BUTTON_INTERFACE

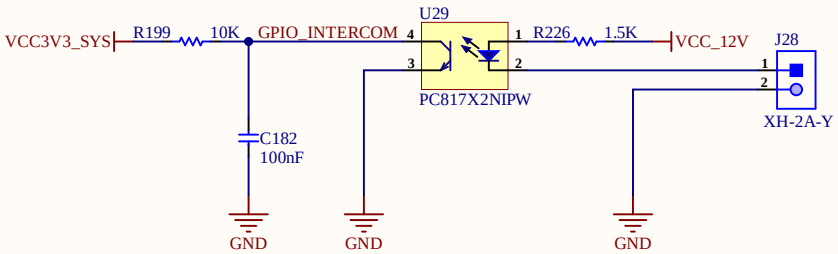
ACTION BUTTONs

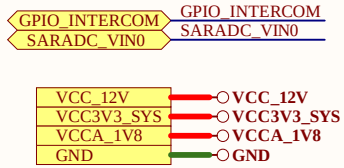


Note:

RECOVERY_Key function:
If SARADC_VIN0=0V
at after power on and reset,
then system will
enter into loader mode.

INTERCOM ACTION BUTTON

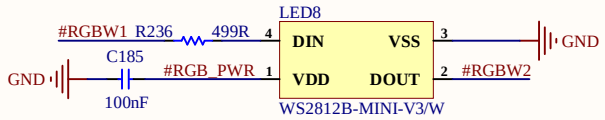


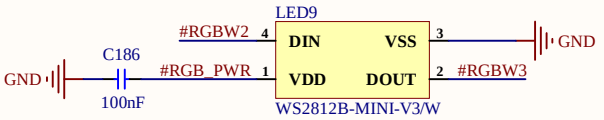


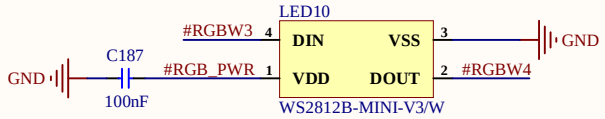
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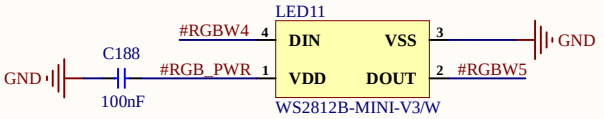
Title: PUSH BUTTONs_INTERFACE.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
Date: 06/09/2024	Page Num: 24
Drawn By: Huzaifa Ali	Page Total: 27

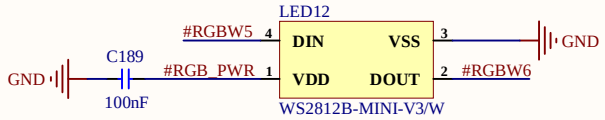
RGB_LEDs_INTERFACE

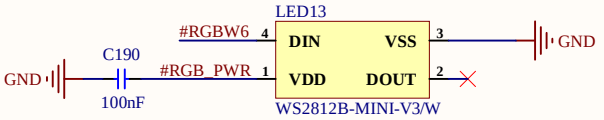


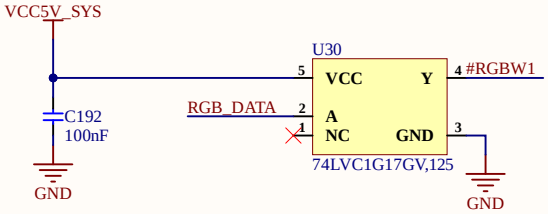


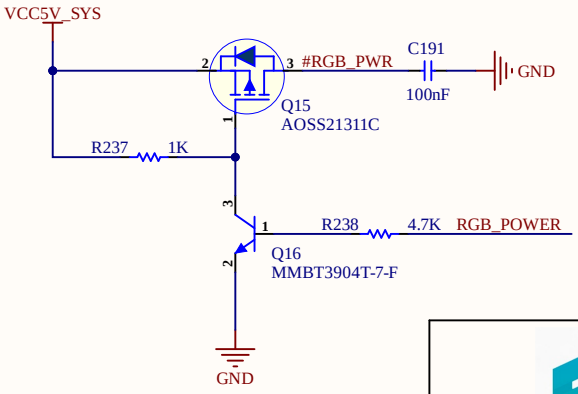
















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Title: RGB_LEDs_INTERFACE.SchDoc		
File: Carrier-Board-v1.1.PrjPcb		
Date: 10/05/2024	Page	Num: 25
Drawn By: Huzaifa Ali	Page	Total: 27

ROCKCHIP_GPIOs_NC PINs

PCIE20_REFCLKP 141
PCIE20_REFCLKN 143
PCIE30_REFCLKP_IN 145
PCIE30_REFCLKN_IN 147
PCIE30_TX0P 150
PCIE30_TX0N 152
PCIE30_TX1P 154
PCIE30_TX1N 156
PCIE30_RX0P 158
PCIE30_RX0N 160
PCIE30_RX1P 162
PCIE30_RX1N 164
PCIE30_PERSTn_M0 185
PCIE30_PERSTn_M1 245
PCIE_CLK_EN 258
PCIE_PWR_EN 151
PCIE30_CLKREQN_M1 135
PCIE30_WAKEn_M1 137

RockChip - PCI-EXPRESS
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J1 Part-R

MIPI_DSI_TX1_D0P 201
MIPI_DSI_TX1_D0N 199
MIPI_DSI_TX1_D1P 205
MIPI_DSI_TX1_D1N 203
MIPI_DSI_TX1_D2P 215
MIPI_DSI_TX1_D2N 213
MIPI_DSI_TX1_D3P 219
MIPI_DSI_TX1_D3N 217
MIPI_DSI_TX1_CLKP 211
MIPI_DSI_TX1_CLKN 209
MIPI_DSI1_BL_EN 41
MIPI_DSI1_RESET 233
MIPI_DSI1_INTERRUPT_IN 173

RockChip - MIPI_DSI1
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J1 Part-O

SATA2_TXP 140
SATA2_TXN 142
SATA2_RXP 144
SATA2_RXN 146
SATA2_ACT_LED 133

RockChip - SATA
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J1 Part-S

MIPI_CSI_RX_D0N 228
MIPI_CSI_RX_D0P 230
MIPI_CSI_RX_D1N 232
MIPI_CSI_RX_D1P 234
MIPI_CSI_RX_D2N 221
MIPI_CSI_RX_D2P 223
MIPI_CSI_RX_D3N 225
MIPI_CSI_RX_D3P 227
MIPI_CSI_RX_CLK0N 236
MIPI_CSI_RX_CLK0P 238
MIPI_CSI_RX_CLK1N 240
MIPI_CSI_RX_CLK1P 242

RockChip - MIPI_CSI
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J1 Part-P

EDP_BL 259 RGB_DATA
EDP_VCC_EN 37
EDP_BL_EN 35
EDP_TX_D0P 96
EDP_TX_D0N 98
EDP_TX_D1P 100
EDP_TX_D1N 102
EDP_TX_D2N 106
EDP_TX_D2P 104
EDP_TX_D3N 110
EDP_TX_D3P 108
EDP_TX_AUXN 107
EDP_TX_AUXP 109

RockChip - LCM_eDP
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J1 Part-T

BAT_ALRT 268 GPIO_INTERCOM
HW_LED 177 HW_LED
PWRON_KEY 43 PWRON_KEY
nPOR_SYS 49 RESET_KEY
EXT_EN 47
4G_PWR_EN 273

RockChip - SYSTEM CTRL
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J1 Part-H

SARADC_VIN0_KEY/RECOVERY 62 SARADC_VIN0
SARADC_VIN3_HW_ID 63 SARADC_VIN3_HW_ID
SARADC_VIN2 61 SARADC_VIN2_LCD_ID
SARADC_VIN4 58 SARADC_VIN4
SARADC_VIN5 60 SARADC_VIN5
SARADC_VIN6 54 SARADC_VIN6
SARADC_VIN7 56 SARADC_VIN7

RockChip - SARADC
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J1 Part-Y

FSPI_CS 53
FSPI_CLK 51
FSPI_D0 57
FSPI_D1 55
FSPI_D2 33
FSPI_D3 52

RockChip - FSPI
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J1 Part-I

CAN1_RX_M1 235
CAN1_TX_M1 237
CAN2_RX_M0 265
CAN2_TX_M0 267

RockChip - CAN
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J1 Part-V

I2S3_SCLK_M0 252 GSM_RST
I2S3_LRCK_M0 253 GSM_PWR_EN
I2S3_SDO_M0 254 GSM_BUCK_EN
I2S3_SDI_M0 256 DEBUG_RSTb

RockChip - I2S
7 of 25

J1 Part-L

SPI1_MOSI_M1 274 I2C_INT
SPI1_CS_M1 241 I2C_RST

RockChip - SPI
21 of 25

J1 Part-U

MULTI_PHY0_REFCLKP 119
MULTI_PHY0_REFCLKN 121
MULTI_PHY1_REFCLKP 125
MULTI_PHY1_REFCLKN 123
PWM0_M0/GPIO0_B7_d 176
RTCIC_INT_L/GPIO0_D3_d 163
CIFCLK_OUT/GPIO4_C0_d 303
GPIO3_A2_d 247
GPIO4_D2_d 261 RGB_POWER
GPIO0_A0_d 170 WIEGAND_W0
GPIO0_C4_d 183 WIEGAND_W1

RockChip - NC_GPIO
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J1 Part-M

SARADC_VIN3_HW_ID C159 1nF GND
SARADC_VIN2_LCD_ID C160 1nF GND
SARADC_VIN0 C161 1nF GND
SARADC_VIN4 C162 1nF GND
SARADC_VIN5 C163 1nF GND
SARADC_VIN6 C164 1nF GND
SARADC_VIN7 C198 1nF GND

I2C_INT I2C_INT
I2C_RST I2C_RST

RGB_DATA RGB_DATA
RGB_POWER RGB_POWER
WIEGAND_W0 WIEGAND_W0
WIEGAND_W1 WIEGAND_W1

SARADC_VIN0 SARADC_VIN0
SARADC_VIN2 SARADC_VIN2_LCD_ID
SARADC_VIN3 SARADC_VIN3_HW_ID
SARADC_VIN4 SARADC_VIN4
SARADC_VIN5 SARADC_VIN5
SARADC_VIN6 SARADC_VIN6
SARADC_VIN7 SARADC_VIN7

RESET_KEY RESET_KEY
PWRON_KEY PWRON_KEY
HW_LED HW_LED

GPIO_INTERCOM GPIO_INTERCOM
DEBUG_RSTb DEBUG_RSTb

GSM_BUCK_EN GSM_BUCK_EN
GSM_RST GSM_RST
GSM_PWR_EN GSM_PWR_EN

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Title: ROCKCHIP_GPIO_NC PINS.SchDoc

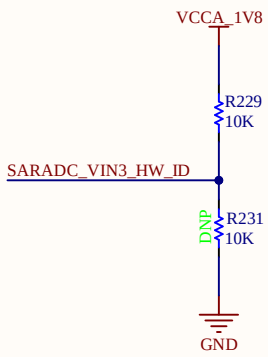
File: Carrier-Board-v1.1.PrjPcb

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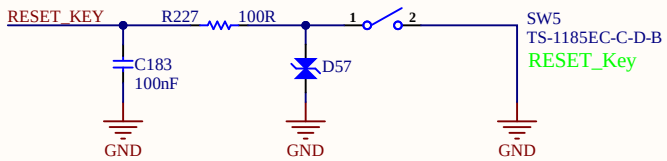
ROCKCHIP SYSTEM CONTROL

SARADC_VIN1	Up Resistance	Down Resistance
HW_ID0	10K	DNP
HW_ID1	10K	110K
HW_ID2	20K	100K
HW_ID3	33K	100K
HW_ID4	18K	36K
HW_ID5	36K	51K
HW_ID6	51K	51K
HW_ID7	51K	36K
HW_ID8	36K	18K
HW_ID9	100K	33K
HW_ID10	100K	20K
HW_ID11	110K	10K
HW_ID12	DNP	10K

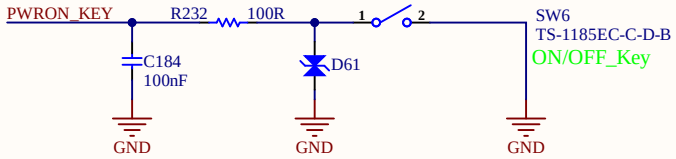
HW_ID0



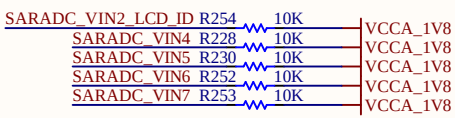
RESET_KEY



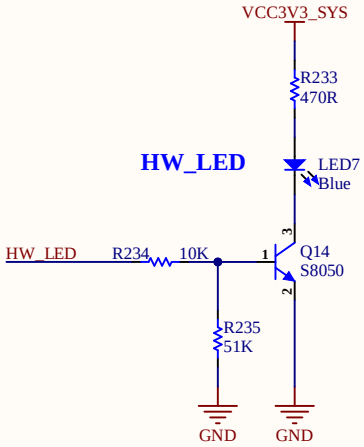
POWERON/OFF_KEY

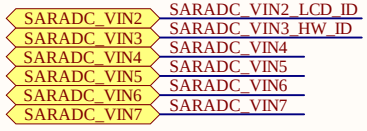


PULL-UP TO VCCA

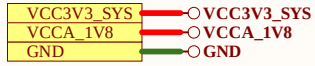


HW_LED









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Title: ROCKCHIP_SYSTEM_CONTROL.SchDoc	
File: Carrier-Board-v1.1.PrjPcb	
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Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.109mm	4.1	
2	L2_GND1	CF-004	0.030mm		
	Dielectric 4	PP-006	0.250mm	4.1	
3	L3_SIG1	CF-004	0.030mm		
	Dielectric 6	PP-006	0.109mm	4.1	
	Dielectric 8	PP-006	0.109mm	4.1	
4	L4_PWR	CF-004	0.030mm		
	Dielectric 1	FR-4	0.250mm	4.8	
5	L5_GND2	CF-004	0.030mm		
	Dielectric 9	PP-006	0.109mm	4.1	
	Dielectric 7	PP-006	0.109mm	4.1	
6	L6_SIG2	CF-004	0.030mm		
	Dielectric 5	PP-006	0.250mm	4.1	
7	L7_GND3	CF-004	0.030mm		
	Dielectric 3	PP-006	0.109mm	4.1	
8	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.674mm		

